A SURJECTIVE MAP ONTO A SOBER SPACE ALONG WHICH SPECIALIZATIONS LIFT DOES NOT DECREASE DIMENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The dimension of a topological space as the supremum of the lengths of chains of irreducible closed subsets, and the lifting property for specializations. Set up in the author's sections on dimension functions and on specialization, before the counted statement. Fixes the quantity to be compared and the hypothesis used to transport chains from the target to the source.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of topological spaces. Suppose that Y is a sober topological space, and f is surjective. If either specializations or generalizations lift along f, then* $\dim(X) \geq \dim(Y)$.

Proof. Assume specializations lift along f. Let $Z_0 \subset Z_1 \subset \ldots Z_e \subset Y$ be a chain of irreducible closed subsets of X. Let $\xi_e \in X$ be a point mapping to the generic point of Z_e. By assumption there exists a specialization $\xi_e \rightsquigarrow \xi_{e-1}$ in X such that ξ_{e-1} maps to the generic point of Z_{e-1}. Continuing in this manner we find a sequence of specializations

$$\xi_e \rightsquigarrow \xi_{e-1} \rightsquigarrow \ldots \rightsquigarrow \xi_0$$

with ξ_i mapping to the generic point of Z_i. This clearly implies the sequence of irreducible closed subsets

$$\overline{\{\xi_0\}} \subset \overline{\{\xi_1\}} \subset \ldots \overline{\{\xi_e\}}$$

is a chain of length e in X. The case when generalizations lift along f is similar. □

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